SKIPS University - School of Computer Science
Master of Science – Information Technology
Semester - II (Batch 2025- 27)
Project Work

Course and Faculty Details:	
Semester	II
Course Code	SOCS010232
Course Name	Data Analysis using Python
Faculty Name	Mr. Dhaval Vaja
Faculty Email ID	dhaval@skipsuniversity.edu.in
General Guidelines:	
Project	Individual Project
Project Submission	Soft Copy only at "eaarjav"
Similarity Index	Not above 20 percent
Total Marks	10
Subject Title of Email	DAP Project_____ Division ___Roll No.__ Name of Student___
Length of the Project	Approximately 10 to 15 pages
Printing	Soft Copy
Font type	Times New Roman
Start Date	02 June 2026
End Date	09 June 2026
Project Guidelines:	
Course Outcomes	CO 2: Apply data cleaning techniques to handle missing values, duplication, and inconsistencies using feature engineering and scaling. CO 3: Illustrate data transformation, aggregation, and summarization through basic visualization techniques using Matplotlib. CO 4: Correlate patterns, relationships, and trends in data using advanced visualization techniques in Matplotlib and Seaborn.
Guidelines for Project Activity	• This is an individual project; all datasets, analyses, and visualizations must be original. • Students must work on a real-world dataset (CSV/Excel/JSON). • Follow the end-to-end data analysis workflow: ○ Data inspection ○ Data cleaning ○ Transformation ○ Visualization ○ Interpretation • Focus first on correct data cleaning and meaningful visualizations before attempting advanced styling. • Use Pandas, Matplotlib, and Seaborn only • Maintain proper file naming and folder structure.

<table>
<tr><td>Project Report (Word Document)</td><td>Report Format
1) Title Page
2) Project Definition and Dataset Description
3) Data Cleaning Techniques Applied
a. Missing values
b. Duplicates
c. Outliers
d. Data transformation
4) Data Analysis and Visualization
a. Descriptive statistics
b. GroupBy operations
c. Basic plots (Line, Bar, Histogram)
d. Advanced plots
5) Insights, Patterns, and Interpretation
6) Conclusion
7) References
Submission of the notebook file with the dataset on the Google Form.</td></tr>
<tr><td>Marking Scheme</td><td><table>
<tr><th>Sr No.</th><th>Component</th><th>CO</th><th>Level</th><th>Weightage</th></tr>
<tr><td>1</td><td>Apply data cleaning techniques on real-world data</td><td>CO2</td><td>L3</td><td>3</td></tr>
<tr><td>2</td><td>Create basic visualizations using Matplotlib</td><td>CO3</td><td>L3</td><td>3</td></tr>
<tr><td>3</td><td>Analyze patterns using advanced visualizations</td><td>CO4</td><td>L4</td><td>4</td></tr>
</table></td></tr>
<tr><td>CO-wise Evaluation</td><td><table>
<tr><th>Sr. No.</th><th>CO</th><th>Weightage</th></tr>
<tr><td>1</td><td>CO2</td><td>3</td></tr>
<tr><td>2</td><td>CO3</td><td>3</td></tr>
<tr><td>3</td><td>CO4</td><td>4</td></tr>
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Signature of Course Faculty: _____ **Signature of Director:** _____

Institute Stamp:

Sr. No.	Roll No.	Student Name	Dataset Title
1	2527001	Anjali Mirwani	Superstore Sales Data
2	2527002	Archi Shah	Online Retail Transactions
3	2527003	Deep Datania	Customer Purchase History
4	2527006	Disha Tanwar	Product Sales Performance
5	2527007	Divyansh Bhavsar	Monthly Revenue Report
6	2527008	Diya Talsaniya	Discount vs Sales Dataset
7	2527009	Gurujyot Singh Kalsi	Store-wise Sales Dataset
8	2527011	Hitanshi Vora	Profit Analysis Dataset
9	2527012	Jancy Mulchandani	Customer Segmentation Data
10	2527013	Jaydip Prajapati	Order Fulfillment Dataset
11	2527014	Jincy Mulchandani	Personal Expense Tracker
12	2527015	Kalpak Shah	Household Budget Dataset
13	2527016	Karina Mulwani	Bank Customer Data
14	2527017	Karunesh Patel	Loan Approval Dataset
15	2527018	Ketankumar Jethva	Credit Card Spending Data
16	2527019	Kevil Shah	Stock Price Data (Single Company)
17	2527021	Mahdiabbas Nayani	Cryptocurrency Price History
18	2527022	Mayur Patel	Insurance Customer Dataset
19	2527023	Meet Ozha	Income vs Expense Dataset
20	2527024	Malwadawala Mo. Faiz	Tax Revenue Dataset
21	2527027	Parth Makwana	Student Performance Dataset
22	2527028	Prachi Shah	Attendance vs Marks Dataset
23	2527031	Raj Rabari	Exam Result Analysis
24	2527032	Shivam Santoki	Online Course Engagement
25	2527033	Tanvi Dave	Placement Statistics Dataset
26	2527034	Vaibhav Thakkar	University Admission Data
27	2527036	Vidhi Kotak	Skill Assessment Dataset
28	2527039	Yogesh Vaishnav	Training Program Data
29	2527040	Aagam Salot	Learning Outcome Survey
30	2527041	Aasana Soni	Faculty Feedback Dataset
31	2527042	Bhakti Vora	Fitness Activity Dataset
32	2527044	Bhavya Parmar	Sleep Pattern Dataset
33	2527047	Dev Joshi	BMI & Lifestyle Dataset
34	2527048	Devrsh Patel	Nutrition & Diet Dataset
35	2527049	Dhavalsinh Parmar	Steps & Calories Dataset
36	2527050	Dhruvi Soni	Wellness Survey Dataset
37	2527051	Dhwani Soni	COVID-19 India Data
38	2527052	Drashti Shah	Hospital Visit Statistics
39	2527053	Harshi Bhavsar	Vaccination Statistics
40	2527054	Harshita Daryani	Life Expectancy Dataset
41	2527057	Jenish Rami	Air Quality Index Data
42	2527059	Kaniz Qureshi	Weather Data (City-wise)
43	2527060	Kartik Vadera	Rainfall Statistics
44	2527061	Mansi Soni	Traffic Accident Data
45	2527062	Mo. Musha Nagori	Road Transport Statistics
46	2527064	Nirav Parmar	Energy Consumption Dataset
47	2527065	Parshwa Bhavin Shah	Water Usage Dataset

Sr. No.	Roll No.	Student Name	Dataset Title
48	2527068	Sam Jaisinghani	Population Census Data
49	2527069	Shreyashi Shah	Crime Statistics Dataset
50	2527070	Tirth Doshi	Public Transport Usage Data
51	2527071	Tirth Goswami	E-Commerce Product Reviews Dataset
52	2527072	Tushar Yogi	Mobile App Store Analytics Dataset
53	2527074	Vinayak Dave	YouTube Channel Performance Dataset
54	2527075	Viraj Dholariya	Movie Ratings and Reviews Dataset
55	2527076	VirajSinh Vaghela	Hotel Booking Demand Dataset
56	2527077	Vraj Vagadia	Restaurant Customer Feedback Dataset